

CONSERVATION PRINCIPLES

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1. INTRODUCTION

We introduce the concepts of conservation of mass, momentum, and energy.

2. CONSERVATION OF ANYTHING

Let the following definitions be made:

- (1) $\mathbb{R}$ is a field of real numbers,
- (2) $\mathbb{E}^3$ is three-dimensional Euclidean space,
- (3) $V \subset \mathbb{E}^3$ is an arbitrary volume of space with infinitesimal volume dV ,
- (4) ∂V is the boundary of the volume V with infinitesimal surface area dA ,
- (5) $d\mathbf{n} \in \mathbb{R}^3$ is the unit outward normal vector of infinitesimal surface area dA on boundary ∂V ,
- (6) $d\mathbf{A}$ be the outward normal with magnitude dA and direction $d\mathbf{n}$,
- (7) $\mathbb{T} = [t_0, t_f] \subset \mathbb{R}$ is the time interval of interest between initial time t_0 and final time t_f . Note, t_0 is not necessarily equal to zero, and the time interval may include negative numbers, in addition to the commonly used positive numbers.

Let $f(\mathbf{x}, t) : V \times \mathbb{T} \mapsto \mathbb{R}$ be a function that describes a material property, defined as item per unit of volume V , varies according to current position $\mathbf{x}$ and time t .

There are two special cases:

- $f = f(t)$ is referred to as **uniform** since f does not vary with position $\mathbf{x}$.
- $f = f(\mathbf{x})$ is referred to as **constant** since f does not vary with time t .

Two equivalent viewpoints are explicated: Eulerian and Lagrangian. With Eulerian, the volume V is fixed in space for all time. With Lagrangian, the volume V moves through space and time.

2.0.1. *Eulerian formulation.* Let point $\mathbf{p}(\mathbf{x}) : V \mapsto \mathbb{R}^3$ be a point fixed in space for all time. Point $\mathbf{p}$ is located by Cartesian coordinates (x, y, z) , thus $\mathbf{p}(\mathbf{x}) = \mathbf{p}(x, y, z)$. Note that $\mathbf{p}$ is not a function of time t . Let the vicinity of volume V be located by point $\mathbf{p}$.

How can property Φ be varied within the volume V ? It can be varied either from *within* the volume or *across* the volume surface, referred to **source** and **flux**, respectively.

- **Source** (or more precisely, point source) says that at point $\mathbf{x}$, the property Φ is created (added) or deleted (subtracted) from a point source or sink, respectively. Let source $\dot{\Phi}_*(\mathbf{x}, t)$ denote the property source/sink rate at point $\mathbf{x}$ and time t with dimensions of property per unit volume per unit time.
- **Flux** describes the property moving across the boundary ∂V . Movement can arise in two ways:
 - (1) Absent of fluid transport: Let $\mathbf{q}$ be the flux of Φ in the absence of fluid transport; for example, with heat conduction.

- (2) Presence of fluid transport: Let $\Phi \mathbf{v}$ be the transport flux of the property because of fluid transport, with units of property per unit area per unit time.

Finally, we state that property Φ is *conserved* because changes in time of Φ across the volume V are caused by either sources or fluxes. Stated mathematically,

$$\frac{d}{dt} \int_V \Phi dV = \underbrace{\int_V \dot{\Phi}_* dV}_{\text{source}} + \underbrace{\left\{ - \int_{\partial V} \mathbf{q} \cdot \mathbf{n} dA - \int_{\partial V} \Phi \mathbf{v} \cdot \mathbf{n} dA \right\}}_{\text{flux}} \quad (1)$$

Note the minus signs for the flux terms are used with the understanding that movement of the property from inside to outside the volume (positive outward flux) *reduces* the time rate of change of the property.

Next, recall the divergence theorem, also known as Gauss's theorem, relates flow (flux) of a vector field through a surface to the behavior of the vector field inside of the surface. That is, for a volume V that is compact and has a piecewise smooth boundary ∂V and unit outward normal $\mathbf{n}$, if $\mathbf{f}$ is a continuously differentiable vector field defined in the vicinity of V , then

$$\int_{\partial V} (\mathbf{f} \cdot \mathbf{n}) dA = \int_V (\nabla_x \cdot \mathbf{f}) dV \quad (2)$$

Use of the divergence theorem used for the flux terms in (1) gives

$$\frac{d}{dt} \int_V \Phi dV = \underbrace{\int_V \dot{\Phi}_* dV}_{\text{source}} - \underbrace{\int_V \nabla_x \cdot (\mathbf{q} + \Phi \mathbf{v}) dV}_{\text{flux}} \quad (3)$$

Because the volume V and surface ∂V are fixed in an inertial reference frame, the time derivative of summed properties equals the sum of local time derivatives of properties, or

$$\frac{d}{dt} \int_V \Phi dV = \int_V \frac{\partial \Phi}{\partial t} dV \quad (4)$$

Use of the foregoing equation in (3) gives

$$\int_V \left(\frac{\partial \Phi}{\partial t} + \nabla_x \cdot (\mathbf{q} + \Phi \mathbf{v}) \right) dV = \underbrace{\int_V \dot{\Phi}_* dV}_{\text{source}} \quad (5)$$

Since the volume V has arbitrary shape and size, the following must also be true, which is the general form for all conservation laws in continuum mechanics,

$$\frac{\partial \Phi}{\partial t} + \nabla_x \cdot (\mathbf{q} + \Phi \mathbf{v}) = \dot{\Phi}_* \quad (6)$$

3. THE MATERIAL DERIVATIVE

Let $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ be the function of a particle (of solid, fluid, or gas) at time t that was located at position $\mathbf{X}$ at time $t = 0$. Let some scalar property Φ (for example, temperature or density) of the fluid be attached the fluid at $\mathbf{X}$ and as this point located by $\mathbf{X}$ moves, the property Φ is invariant in time. This means

$$\left. \frac{\partial \Phi}{\partial t} \right|_{\mathbf{X}} = 0. \quad (7)$$

When a property Φ is invariant in time for a particle $\mathbf{X}$ of the fluid, the invariant scalar is *material*.

The Eulerian point of view of this same property Φ proceeds by selecting a point $\mathbf{x}$ that is fixed in space, and expressing the material invariance in terms of properties of the fluid *at* point $\mathbf{x}$. In general, the fluid will

be moving at this point $\mathbf{x}$, so the velocity $\mathbf{v}$ of the fluid at this point enters through the chain rule. That is

$$\left. \frac{\partial \Phi(\boldsymbol{\varphi}(\mathbf{X}, t), t)}{\partial t} \right|_{\mathbf{X}} \stackrel{(7)}{=} 0 = \left. \frac{\partial \Phi}{\partial \varphi_i} \right|_t \frac{\partial \varphi_i}{\partial t} \Big|_{\mathbf{X}} + \left. \frac{\partial \Phi}{\partial t} \right|_{\boldsymbol{\varphi}} = \nabla_x \Phi \cdot \mathbf{v} + \Phi_{,t}. \quad (8)$$

This gives rise to the notation of **material derivative**, written with capital D as

$$\frac{D(\bullet)}{Dt} \triangleq \frac{\partial(\bullet)}{\partial t} + \mathbf{v} \cdot \nabla_x(\bullet). \quad (9)$$

4. CONSERVATION OF MASS

Conservation of mass is obtained from (6) with the following substitutions:

- $\Phi = \rho(\mathbf{x}, t)$ is the mass per unit volume (or density),
- $\mathbf{q} = \mathbf{0}$ because flux of mass can only occur with transport, and
- $\dot{\Phi}_* = 0$ since mass can be neither created or destroyed.

This gives conservation of mass, also called the **continuity equation**

$$\frac{\partial \rho}{\partial t} + \nabla_x \cdot (\rho \mathbf{v}) = 0. \quad (10)$$